Paper 8.50 - Lattice Closure Claim Contract

CQE-paper-08.50
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-08.50.md

Paper 8.50 - Lattice Closure Claim Contract

Purpose

Paper 8.50 defines what counts as a valid lattice closure claim. It separates verified local code/lattice facts from unproved global landing claims.

Admitted Paper 8 Claims

The following claims are admitted from Paper 8:

```
the local chain (1,3,7,8,24) passes verifier checks
dimension 7 passes Fano/Hamming identity checks
dimension 8 passes extended Hamming/E8 seed checks
dimension 24 supplies Golay ingredients and 24 = 3 x 8 geometry
the powered terminal satisfies 8 x 9 = 72 with K = 9
Gamma72 transport round trips are exact for checked payloads
Leech landing and Gamma72 polarization remain unproved by this receipt
```

Claim Requirements

Any later paper using Paper 8 must state:

```
which rung is being imported
which verifier or receipt proves the rung
whether the claim is local transport, ingredient evidence, or global landing
whether any Leech/Gamma72/polarization claim has a separate receipt
```

Linked Receipt

The minimum receipt link for Paper 8 is:

```
paper: CQE-paper-08
theorem: Local lattice closure template
receipt: production/formal-papers/CQE-paper-08/lattice_closure_template_receipt.json
status: pass
```

The receipt is sufficient for the local closure scaffold. It is not sufficient by itself for rootless Leech landing, Gamma72 polarization, cold-start fingerprint universality, or uniqueness of all possible closure chains.

Boundary Failures

The following are boundary failures:

```
claiming Golay ingredients prove the rootless Leech landing
claiming three-sheet Gamma72 transport proves polarization
claiming this chain is the only possible closure chain
importing lattice terms without naming the checked rung
using a continuous bridge from Paper 7 as closure proof without this receipt
```

Boundary failures become audit boundaries or are rejected.

Hidden-Guess Contract

When hidden-guess diagnostics are enabled, the answer must be recorded before the receipt is revealed. The guess record should include:

```
prompt
predicted rung result or scope status
revealed receipt
match/mismatch
next audit boundary if mismatch
```

The Leech and Gamma72 overclaim prompts are required because their correct answer is negative in this paper.

Conclusion

Paper 8.50 lets later papers import the lattice closure scaffold honestly. It keeps the verified local rungs useful while preserving global landings as separate proof obligations.